

Course Code : ICE 3146 **Course Title :** Microprocessor and Interfacing Laboratory
Lab Task No. : 05
Lab Task Topic: Interfacing and Operating a Seven Segment Display using the 8086 Microprocessor kit in Kit Modes and PC Modes.

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Title: Interfacing and Operating a Seven Segment Display using the 8086 Microprocessor kit in Kit Modes and PC Modes.

Abstract:

This experiment was performed using the Intel 8086 Microprocessor Trainer Kit to study the interfacing and operation of a seven-segment display in Kit and PC modes. The main objective was to understand how the 8086 microprocessor sends data to an external output device through the 8255 Programmable Peripheral Interface (PPI) and how the display can be controlled by assembly language instructions. In this experiment, the control port of the 8255 was initialized first, and then predefined segment codes were sent sequentially to Port A. As a result, the display showed the sequence 11, 9, 4, 2, and 1. This experiment helped to understand port addressing, PPI initialization, display encoding, the OUT instruction, and practical hardware interfacing using the 8086-microprocessor.

Introduction (Theory):

The Intel 8086 is a 16-bit microprocessor that communicates with external hardware devices through Input/Output (I/O) ports. Devices such as LEDs and seven-segment displays are common output peripherals, and connecting these devices with the microprocessor is known as interfacing. In assembly language, the OUT instruction is used to transfer data from a register to an output port so that the connected hardware can respond accordingly.

A seven-segment display is an electronic output device used to represent decimal digits. It consists of seven individual segments labeled a, b, c, d, e, f, and g. Different combinations of ON and OFF segments produce different numbers. In this experiment, the seven-segment display was driven through the 8255 Programmable Peripheral Interface, which provides programmable ports for communication between the 8086 microprocessor and the display hardware.

The control word 80H was sent to the control port to configure the 8255 in output mode. After that, segment codes were sent one after another to Port A. Each code corresponded to a particular displayed output. The observed sequence in this experiment was 11, 9, 4, 2, and 1. The experiment demonstrates practical use of port initialization, direct data output, and hardware operation through the 8086 trainer kit. In Kit Mode the output can be checked directly from the trainer environment, while in PC Mode the same logic can be assembled, downloaded, and executed through the computer interface.

Material:

Hardware	Software
1. Intel 8086 Microprocessor Trainer Kit	1. Microsoft Macro Assembler (MASM)
2. Seven-Segment Display Interface Module	2. MDA-8086 / MDA-WinIDE8086
3. Personal Computer (PC)	3. Text Editor (Notepad)
4. Connecting Cable / Interface Cable	4. Command Prompt (CMD)

Method:

Code:

```
CODE SEGMENT
    ASSUME CS: CODE
    ORG 1000H
    Con_c EQU 1FH
    Port_a EQU 19H
    MOV AL, 10000000B
    OUT Con_c, AL
    MOV AL, 11001001B
    OUT port_a, AL
    MOV AL, 10010000B
    OUT port_a, AL
    MOV AL, 10011001B
    OUT port_a, AL
    MOV AL, 10100100B
    OUT port_a, AL
    MOV AL, 11111001B
    OUT port_a, AL
CODE ENDS
END
```

Execution (Kit Mode):

1. First, the assembly language program was written in Notepad and saved as .asm file.
2. Machine code execution using CMD-

```
C:\MDA\8086\Asm8086>MASM.exe SUM26A14.asm
```

```
Object filename [SUM26_1.OBJ]: SUM26A14
```

```
Source listing [NUL.LST]: SUM26A14
```

```
Cross-reference [NUL.CRF]:
```

```
    0 Warning Errors
```

```
    0 Severe Errors
```

3. The 8086-microprocessor trainer kit was turned ON and reset.
4. The data entry mode was selected by pressing the DA key.
5. The Machine Code was entered manually using the Number Pad from the Kit.
6. The + key was pressed to move to the next memory location.
7. After entering all machine codes, the REG key was pressed to check the register values.
8. The program was executed step by step by pressing the STP key. After each STP key press, the Seven Segment Display shows the numbers according the code.

Execution (PC Mode):

1. First, the assembly language program was written in Notepad and saved as .asm file.
2. Machine code execution using CMD-

```
C:\MDA\8086\Asm8086>MASM.exe SUM26A14.asm
```

Object filename [SUM26_1.OBJ]: SUM26A14

Source listing [NUL.LST]: SUM26A14

Cross-reference [NUL.CRF]:

0 Warning Errors

0 Severe Errors

9. The MDA (Microprocessor Development and Analysis) software was opened.
10. The SUM26A13.asm file was loaded using the CMD panel (LO).
11. The program was assembled and linked to generate the .ABS (Absolute) file using the Assemble & Link option.
12. The generated .ABS file was loaded into the simulator.
13. The program was executed using the Run command.
14. The output and register values were observed through the simulator window and the Seven Segment Display shows the numbers according the code.

Result:

Result for Kit Mode:

		
Initial condition of Display	Output for 11001001B	Output for 10010000B
		

Output for 10011001B	Output for 10100100B	Output for 11111001B
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Fig 1: Seven Segment Display outputs for different Sequence of Code.

Result for PC Mode:

```

** 8086 Monitor 9.5 **
** WWW.MIDASENG.COM **
8086 >LOD186 SUM26A14.ASM
    Down load start !!
:18100000B080E61FB0C9E619B090E619B099E619B0A4E619B0F9
:00
    OK Completed !!

8086 >G
Execute Address = 0000:1000
Divide 0 interrupt = 0000:101E

    AX=00F9  BX=0000  CX=0000  DX=0000
    SP=053C  BP=0000  SI=0000  DI=0000
    DS=0000  ES=0000  SS=0000  CS=0000
    IP=101E  FL=0044  = . . . . . Z . P .

```

Fig 2: Program execution in PC Mode using 8086 monitor terminal.

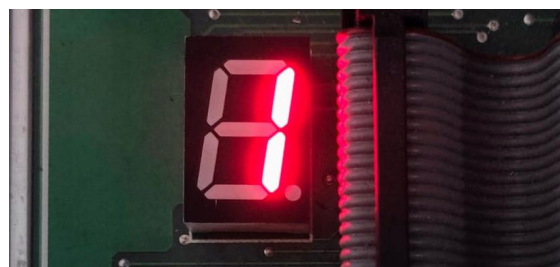


Fig 3: LED output for PC Mode.

Discussion:

In this experiment, we studied the interfacing and operation of a seven-segment display with the 8086 microprocessor kit in both Kit Mode and PC Mode. The control port 1FH was initialized first, and then different binary values were sent to port 19H using the OUT instruction. Each binary value produced a different output on the seven-segment display. In Kit Mode, the machine codes were entered manually and executed from the trainer kit. In PC Mode, the program was loaded and executed through the MDA 8086 monitor software. Through this experiment, we understood how the 8086 microprocessor sends data to an output port to control a display device.